DAKINI AI | ENTERPRISE SALES PLATFORM

Complete Database Schema, RBAC Permissions & Entity Relationships Specification

Formal System Architecture Reference covering 32 Relational Tables, Multi-Tenant Security, 3-Tier Role Governance (Owner, Manager, Salesperson), and 1:N / N:M Relationship Topology.

Target Engine: PostgreSQL 14+ / Supabase	**Architecture:** 3-Tier Multi-Tenant RBAC	**Security Mode:** Strict Row-Level Security (RLS)
Tables Cataloged: 32 Database Entities	**Canonical Roles:** Owner, Manager, Salesperson	**Specification Version:** v2.1 Hardened

1. Executive Overview & 3-Tier Role Model

Dakini Sales CRM operates on a zero-trust multi-tenant architecture with cryptographic tenant isolation. Access control is governed by three canonical roles defined at Layer 3 and enforced via database Row Level Security (RLS):

Role	Canonical Identifier	Authorization Scope & Core Capabilities	Administrative Boundaries
Owner / Director	`owner`	Full organization-wide authority. Can access, create, modify, and delete all accounts, deals, contacts, leads, and custom reports. Sole role permitted to delete accounts and perform system admin.	Unrestricted within tenant. Cross-tenant access strictly prevented.
Sales Manager	`manager`	Team-level operational supervisor. Can view, manage, and assign leads/deals belonging to their active sales team. Can create accounts/contacts and set quotas for direct-report sales reps.	Strictly forbidden from deleting accounts, modifying Owners, or viewing other pods' records.
Salesperson (User)	`salesperson`	Direct individual contributor. Can access only records explicitly assigned to or owned by them. Can create leads, accounts, contacts, and deals. Can move deal pipeline stages.	Strictly forbidden from deleting accounts, contacts, deals, or leads. Forbidden from unassigned pool.

2. Full Database Schema Catalog (32 Tables)

Below is the complete database dictionary cataloging all 32 relational tables partitioned into 7 domain groups:

Domain 1: Multi-Tenancy & Identity Access Management (IAM)

Table: 1. organizations — *Tenant root entity isolating all CRM data*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Unique tenant organization identifier
`name`	VARCHAR(100)	NOT NULL	Enterprise corporate name
`slug`	VARCHAR(100)	NOT NULL, UNIQUE	Unique URL slug
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Last update timestamp

Table: 2. roles — *Tenant capability bundles and role assignments*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Unique role identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Parent tenant
`name`	VARCHAR(100)	NOT NULL	Role designation (owner, manager, salesperson)
`description`	TEXT	NULL	Role permissions summary
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Last update timestamp

Table: 3. permissions — *Atomic system actions and resource scopes*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Permission identifier
`resource`	VARCHAR(50)	NOT NULL	Resource type (leads, deals, accounts)
`action`	VARCHAR(50)	NOT NULL	Allowed operation (create, read, update, delete)
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Last update timestamp

Table: 4. role_permissions — *Join table mapping roles to atomic permissions*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Mapping identifier
`role_id`	UUID	**NOT NULL, FK -> roles(id) CASCADE**	Associated role
`permission_id`	UUID	**NOT NULL, FK -> permissions(id) CASCADE**	Granted permission
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Assignment timestamp

Table: 5. users — *System employees, managers, and representatives*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	User identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant identifier
`role_id`	UUID	**NULL, FK -> roles(id) SET NULL**	Assigned role record
`manager_id`	UUID	**NULL, FK -> users(id) SET NULL**	Direct line supervisor
`role_name`	VARCHAR(50)	NOT NULL, DEFAULT 'salesperson'	Canonical role: 'owner', 'manager', 'salesperson'
`login_id`	VARCHAR(100)	NULL	Case-insensitive normalized login name
`email`	VARCHAR(150)	NOT NULL, UNIQUE	Work email address
`name`	VARCHAR(100)	NOT NULL	Display name
`hashed_password`	VARCHAR(255)	NULL	Bcrypt hash password
`status`	VARCHAR(20)	NOT NULL, DEFAULT 'active'	'active', 'inactive', 'suspended'

must_change_password	BOOLEAN	NOT NULL, DEFAULT FALSE	Forced reset flag
password_changed_at	TIMESTAMPTZ	NULL	Timestamp of last password change
token_version	INTEGER	NOT NULL, DEFAULT 1	Token revocation version counter
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
deleted_at	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 6. teams — *Sales pods and regional operating units*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Team identifier
organization_id	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Parent tenant
manager_id	UUID	**NULL, FK -> users(id) SET NULL**	Assigned team manager
name	VARCHAR(100)	NOT NULL	Team designation
description	TEXT	NULL	Territory or objective
status	VARCHAR(20)	NOT NULL, DEFAULT 'active'	'active', 'archived'
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
deleted_at	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 7. team_members — *Sales rep team membership mapping*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Membership identifier
team_id	UUID	**NOT NULL, FK -> teams(id) CASCADE**	Team reference
user_id	UUID	**NOT NULL, FK -> users(id) CASCADE**	Sales representative
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Active assignment flag
joined_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Membership start date

Domain 2: Sales Acquisition, Leads & Marketing

Table: 8. campaigns — *Marketing roadshows, webinars, and inbound programs*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Campaign identifier
organization_id	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
name	VARCHAR(100)	NOT NULL	Campaign title
type	VARCHAR(50)	NOT NULL	'Roadshow & In-Person', 'Digital & Social'
status	VARCHAR(20)	NOT NULL, DEFAULT 'planning'	'planning', 'active', 'completed'
start_date	DATE	NULL	Launch date
end_date	DATE	NULL	End date
budget	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Allocated budget in USD
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
deleted_at	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 9. leads — *Inbound demo requests and prospective buyers*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Lead identifier
organization_id	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
assigned_to	UUID	**NULL, FK -> users(id) SET NULL**	Assigned sales rep
campaign_id	UUID	**NULL, FK -> campaigns(id) SET NULL**	Originating campaign

`name`	VARCHAR(100)	NOT NULL	Prospect name
`email`	VARCHAR(150)	NULL	Contact email
`phone`	VARCHAR(20)	NULL	Phone number
`industry`	VARCHAR(50)	NULL	Industry vertical
`source`	VARCHAR(50)	NULL	'Website', 'Inbound', 'Referral'
`status`	VARCHAR(20)	NOT NULL, DEFAULT 'New'	'New', 'Contacted', 'Qualified', 'Converted'
`score`	INTEGER	NOT NULL, DEFAULT 0	AI/ICP qualification score
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
`deleted_at`	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 10. lead_signals — *AI and behavioural intent signals*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Signal identifier
`lead_id`	UUID	**NOT NULL, FK -> leads(id) CASCADE**	Target lead
`signal_type`	VARCHAR(50)	NOT NULL	'demo_request', 'pricing_visit'
`signal_value`	TEXT	NULL	Event payload/context
`score`	INTEGER	NOT NULL, DEFAULT 0	Points contributed
`source`	VARCHAR(50)	NULL	'clearbit', 'google_tag'
`occurred_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Event occurrence time
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Ingestion timestamp

Table: 11. campaign_members — *Leads and contacts enrolled in campaigns*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Membership identifier
`campaign_id`	UUID	**NOT NULL, FK -> campaigns(id) CASCADE**	Parent campaign
`lead_id`	UUID	**NULL, FK -> leads(id) CASCADE**	Enrolled lead
`contact_id`	UUID	**NULL, FK -> contacts(id) CASCADE**	Enrolled contact
`status`	VARCHAR(20)	NOT NULL, DEFAULT 'sent'	'sent', 'opened', 'clicked'
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp

Table: 12. campaign_responses — *Customer feedback and engagement replies*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Response identifier
`campaign_member_id`	UUID	**NOT NULL, FK -> campaign_members(id) CASCADE**	Member reference
`response_type`	VARCHAR(50)	NOT NULL	'demo_booked', 'replied'
`response_date`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Response timestamp
`notes`	TEXT	NULL	Transcript text
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp

Domain 3: Customer Accounts & Stakeholder Contacts

Table: 13. accounts — *Enterprise client organizations and companies*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Account identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`name`	VARCHAR(100)	NOT NULL	Enterprise legal name
`industry`	VARCHAR(50)	NULL	Industry (e.g. Logistics & Tech)

website	VARCHAR(150)	NULL	Corporate URL
phone	VARCHAR(20)	NULL	Corporate phone
address	TEXT	NULL	Headquarters location
status	VARCHAR(20)	NOT NULL, DEFAULT 'active'	'active', 'prospect', 'churned'
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
deleted_at	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 14. contacts — *Buyers, VPs, and decision makers*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Contact identifier
account_id	UUID	**NOT NULL, FK -> accounts(id) CASCADE**	Parent enterprise
first_name	VARCHAR(50)	NOT NULL	First name
last_name	VARCHAR(50)	NOT NULL	Surname
email	VARCHAR(150)	NULL	Direct email
phone	VARCHAR(20)	NULL	Direct phone
title	VARCHAR(100)	NULL	Executive title
department	VARCHAR(50)	NULL	Department
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
deleted_at	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 15. lead_conversions — *Historical audit records of lead conversions*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Conversion identifier
lead_id	UUID	**NOT NULL, FK -> leads(id) CASCADE**	Converted lead
account_id	UUID	**NOT NULL, FK -> accounts(id) CASCADE**	Target account
opportunity_id	UUID	**NULL, FK -> opportunities(id) SET NULL**	Resulting opportunity
conversion_type	VARCHAR(50)	NOT NULL, DEFAULT 'standard'	Conversion mode
conversion_date	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Conversion timestamp
notes	TEXT	NULL	Conversion notes

Domain 4: Pipeline, Deals, Commerce & Quotas

Table: 16. opportunities — *Long-term sales pipeline opportunities*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Opportunity identifier
organization_id	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
account_id	UUID	**NOT NULL, FK -> accounts(id) CASCADE**	Target account
contact_id	UUID	**NULL, FK -> contacts(id) SET NULL**	Key stakeholder
owner_id	UUID	**NULL, FK -> users(id) SET NULL**	Assigned sales owner
name	VARCHAR(100)	NOT NULL	Deal title
amount	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Deal value in USD
stage	VARCHAR(50)	NOT NULL, DEFAULT 'discovery'	'discovery', 'qualified', 'proposal', 'negotiation', 'won'
probability	INTEGER	NOT NULL, DEFAULT 10	Win probability (0-100)
expected_close_date	DATE	NULL	Target closing date
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp

`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
`deleted_at`	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 17. deals — *Active Kanban execution records*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Deal identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`account_id`	UUID	**NOT NULL, FK -> accounts(id) CASCADE**	Associated account
`opportunity_id`	UUID	**NULL, FK -> opportunities(id) SET NULL**	Linked opportunity
`owner_id`	UUID	**NULL, FK -> users(id) SET NULL**	Assigned sales rep
`name`	VARCHAR(100)	NOT NULL	Deal name
`amount`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Deal ARR value
`currency`	VARCHAR(3)	NOT NULL, DEFAULT 'USD'	Currency code
`stage`	VARCHAR(50)	NOT NULL, DEFAULT 'Qualified'	'Qualified', 'Proposal', 'Negotiation', 'Won', 'Lost'
`status`	VARCHAR(20)	NOT NULL, DEFAULT 'open'	'open', 'won', 'lost'
`close_date`	DATE	NULL	Closing date
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
`deleted_at`	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 18. products — *Central price catalog and sellable units*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Product identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`name`	VARCHAR(100)	NOT NULL	Product name
`sku`	VARCHAR(50)	NOT NULL	Stock unit code
`category`	VARCHAR(50)	NULL	Category (SaaS, Hardware)
`description`	TEXT	NULL	Product details
`price`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Base price
`currency`	VARCHAR(3)	NOT NULL, DEFAULT 'USD'	Currency
`is_active`	BOOLEAN	NOT NULL, DEFAULT TRUE	Active in catalog
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
`deleted_at`	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 19. deal_products — *Line-item products attached to sales deals*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Line-item identifier
`deal_id`	UUID	**NOT NULL, FK -> deals(id) CASCADE**	Linked deal
`product_id`	UUID	**NOT NULL, FK -> products(id) RESTRICT**	Catalog item
`quantity`	INTEGER	NOT NULL, DEFAULT 1	Units sold
`unit_price`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Price per unit
`discount`	DECIMAL(5,2)	NOT NULL, DEFAULT 0.00	Discount applied (%)
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp

Table: 20. sales_targets — *Quota allocations assigned to sales reps and teams*

Column Name	Data Type	Constraints & Keys	Description

`id`	UUID	PRIMARY KEY, gen_random_uuid()	Target identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`assigned_to_user_id`	UUID	**NOT NULL, FK -> users(id) CASCADE**	Recipient sales rep
`assigned_by_user_id`	UUID	**NULL, FK -> users(id) RESTRICT**	Assigning manager/director
`team_id`	UUID	**NULL, FK -> teams(id) CASCADE**	Team reference
`period_type`	VARCHAR(20)	NOT NULL, DEFAULT 'monthly'	'monthly', 'quarterly'
`period_start`	DATE	NOT NULL	Quota start date
`period_end`	DATE	NOT NULL	Quota end date
`target_amount`	DECIMAL(14,2)	NOT NULL, CHECK > 0	Target revenue in USD
`currency`	VARCHAR(3)	NOT NULL, DEFAULT 'USD'	Currency
`status`	VARCHAR(20)	NOT NULL, DEFAULT 'active'	'active', 'completed'
`notes`	TEXT	NULL	Quota allocation notes
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp

Domain 5: Engagement, Activities & Notes

Table: 21. activities — *Tasks, calendar meetings, and customer calls*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Activity identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`account_id`	UUID	**NULL, FK -> accounts(id) CASCADE**	Linked account
`contact_id`	UUID	**NULL, FK -> contacts(id) CASCADE**	Linked contact
`opportunity_id`	UUID	**NULL, FK -> opportunities(id) CASCADE**	Linked opportunity
`user_id`	UUID	**NULL, FK -> users(id) SET NULL**	Assigned rep
`type`	VARCHAR(50)	NOT NULL	'task', 'call', 'meeting', 'email'
`subject`	VARCHAR(150)	NOT NULL	Activity title
`description`	TEXT	NULL	Activity notes
`start_at`	TIMESTAMPTZ	NULL	Start time
`end_at`	TIMESTAMPTZ	NULL	End time
`timezone`	VARCHAR(50)	NOT NULL, DEFAULT 'UTC'	Timezone
`status`	VARCHAR(20)	NOT NULL, DEFAULT 'scheduled'	'scheduled', 'completed', 'canceled'
`priority`	VARCHAR(20)	NOT NULL, DEFAULT 'medium'	'low', 'medium', 'high', 'urgent'
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp

Table: 22. activity_participants — *Attendees on meetings and activities*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Participant identifier
`activity_id`	UUID	**NOT NULL, FK -> activities(id) CASCADE**	Parent activity
`user_id`	UUID	**NULL, FK -> users(id) CASCADE**	Internal attendee
`contact_id`	UUID	**NULL, FK -> contacts(id) CASCADE**	External attendee
`response_status`	VARCHAR(20)	NOT NULL, DEFAULT 'pending'	'pending', 'accepted'
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp

Table: 23. notes — *Polymorphic notes on Leads, Deals, Accounts, Contacts*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Note identifier

`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`user_id`	UUID	**NULL, FK -> users(id) SET NULL**	Authoring user
`related_to_type`	VARCHAR(50)	NOT NULL	Entity type (lead, deal, account)
`related_to_id`	UUID	NOT NULL	Target entity ID
`content`	TEXT	NOT NULL	Note text
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp

Domain 6: Billing, Invoices & Payments

Table: 24. invoices — *Commercial customer billing invoices*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Invoice identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`account_id`	UUID	**NOT NULL, FK -> accounts(id) CASCADE**	Billed account
`deal_id`	UUID	**NULL, FK -> deals(id) SET NULL**	Source deal
`invoice_number`	VARCHAR(50)	NOT NULL, UNIQUE	Invoice code (INV-2026-001)
`issue_date`	DATE	NOT NULL	Issue date
`due_date`	DATE	NOT NULL	Due date
`subtotal`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Net amount
`tax_amount`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Tax amount
`total_amount`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Total payable
`currency`	VARCHAR(3)	NOT NULL, DEFAULT 'USD'	Currency
`status`	VARCHAR(20)	NOT NULL, DEFAULT 'draft'	'draft', 'sent', 'paid', 'overdue'
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp

Table: 25. invoice_items — *Line-item products on invoices*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Item identifier
`invoice_id`	UUID	**NOT NULL, FK -> invoices(id) CASCADE**	Parent invoice
`product_id`	UUID	**NULL, FK -> products(id) SET NULL**	Catalog item
`description`	TEXT	NOT NULL	Item description
`quantity`	INTEGER	NOT NULL, DEFAULT 1	Quantity
`unit_price`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Unit price
`discount`	DECIMAL(5,2)	NOT NULL, DEFAULT 0.00	Discount
`tax`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Tax amount
`subtotal`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Subtotal
`total`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Total price
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp

Table: 26. payments — *Customer payment receipts and remittances*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Payment identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`invoice_id`	UUID	**NOT NULL, FK -> invoices(id) CASCADE**	Settled invoice
`amount`	DECIMAL(12,2)	NOT NULL, DEFAULT 0.00	Payment amount

payment_method	VARCHAR(50)	NOT NULL	'wire_transfer', 'credit_card'
transaction_id	VARCHAR(100)	NULL	Bank reference
payment_date	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Payment timestamp
status	VARCHAR(20)	NOT NULL, DEFAULT 'completed'	'completed', 'failed'
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp

Domain 7: Content, Auditing & Reliability

Table: 27. documents — *Proposals, NDAs, and contract files*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Document identifier
organization_id	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
uploaded_by	UUID	**NULL, FK -> users(id) SET NULL**	Author rep
name	VARCHAR(255)	NOT NULL	Document title
description	TEXT	NULL	Description
document_type	VARCHAR(50)	NOT NULL, DEFAULT 'other'	'contract', 'proposal', 'nda'
file_url	VARCHAR(255)	NULL	Presigned file URL
mime_type	VARCHAR(100)	NULL	MIME type (application/pdf)
file_size	BIGINT	NOT NULL, DEFAULT 0	File size in bytes
storage_path	TEXT	NOT NULL	S3/Supabase storage key
status	VARCHAR(20)	NOT NULL, DEFAULT 'draft'	'draft', 'final', 'expired'
expires_at	DATE	NULL	Contract expiry
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Update timestamp
deleted_at	TIMESTAMPTZ	NULL	Soft delete timestamp

Table: 28. document_relations — *Polymorphic associations to Deals or Accounts*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Relation identifier
document_id	UUID	**NOT NULL, FK -> documents(id) CASCADE**	Parent document
related_to_type	VARCHAR(50)	NOT NULL	'deal', 'account', 'lead'
related_to_id	UUID	NOT NULL	Target entity ID
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp

Table: 29. document_versions — *Version revision history for contracts*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Version identifier
document_id	UUID	**NOT NULL, FK -> documents(id) CASCADE**	Parent document
version_number	INTEGER	NOT NULL, DEFAULT 1	Revision counter
storage_path	TEXT	NOT NULL	Storage key for revision
file_size	BIGINT	NOT NULL, DEFAULT 0	File size in bytes
uploaded_by	UUID	**NULL, FK -> users(id) SET NULL**	Revision author
change_note	TEXT	NULL	Release notes
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp

Table: 30. audit_logs — *Immutable append-only compliance audit trail*

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY, gen_random_uuid()	Log identifier

`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`user_id`	UUID	**NULL, FK -> users(id) SET NULL**	Actor rep
`action`	VARCHAR(50)	NOT NULL	'CREATE', 'UPDATE', 'DELETE'
`resource_type`	VARCHAR(50)	NOT NULL	'lead', 'deal', 'account'
`resource_id`	UUID	NOT NULL	Affected record ID
`old_values`	JSONB	NULL	Previous record state
`new_values`	JSONB	NULL	New record state
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Event timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Timestamp

Table: 31. outbox_events — *Transactional outbox for event publishing*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Event identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`event_type`	VARCHAR(100)	NOT NULL	'deal.won', 'lead.converted'
`payload`	JSONB	NOT NULL	Event JSON
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Event generation time
`published_at`	TIMESTAMPTZ	NULL	Broker publish timestamp

Table: 32. idempotency_keys — *Gateway request deduplication store*

Column Name	Data Type	Constraints & Keys	Description
`id`	UUID	PRIMARY KEY, gen_random_uuid()	Key identifier
`organization_id`	UUID	**NOT NULL, FK -> organizations(id) CASCADE**	Tenant owner
`key`	VARCHAR(255)	NOT NULL	Client Idempotency-Key
`request_hash`	VARCHAR(64)	NOT NULL	SHA-256 payload hash
`response`	JSONB	NOT NULL	Cached HTTP response
`expires_at`	TIMESTAMPTZ	NOT NULL	Expiry timestamp
`created_at`	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Creation timestamp

3. Role Permissions Matrix by CRM Module

The following matrix summarizes the exact operational capabilities across all CRM functional modules for each role:

Module / Action	Owner (Director)	Manager	Salesperson (User)	Enforcement Rule & Policy
Org Settings & Admin	**Full Access**	**Forbidden (403)**	**Forbidden (403)**	Strictly Owner only
Export Data (Org-wide)	**Full CSV/Excel**	**Forbidden**	**Forbidden**	check_can_export_data
Export Data (Team Scope)	**Full Access**	**Allowed (Team only)**	**Forbidden (403)**	Team-level export only
View Full Audit Logs	**All Events**	**Forbidden**	**Forbidden**	Owner read-only
Create Users / Invite Reps	**Any Role**	**Forbidden**	**Forbidden**	check_can_manage_user
Create / Delete Teams	**Full Access**	**Forbidden**	**Forbidden**	check_can_manage_team
Assign Reps to Teams	**Any Team**	**Own Team Only**	**Forbidden**	check_can_manage_team_members
Reset User Password	**Any User**	**Team Reps Only**	**Forbidden**	check_can_reset_password
Set Sales Quotas	**Any Manager/Rep**	**Team Reps Only**	**Forbidden (403)**	check_can_manage_sales_target
View Sales Targets	**Org-wide**	**Team + Own**	Own Quota Only	Target user filtering
View Leads	**All Leads**	**Team + Own**	Assigned Only	check_record_access
Unassigned Leads Pool	**Full Access**	**Authorized Mgrs**	**Forbidden (403)**	check_can_access_unassigned_leads
Create Lead	**Allowed**	**Allowed**	**Allowed**	Self or team assignment
Assign Lead to Rep	**Any Rep**	**Team Reps Only**	**Forbidden**	Scoped by team ID
Convert Lead	**Any Lead**	**Team Leads**	**Assigned Leads**	Creates Account, Contact & Deal
Delete Lead	**Any Lead**	**Team Leads Only**	**Forbidden (403)**	check_can_delete_lead
View Accounts	**All Accounts**	**Team Accounts**	Assigned Accounts	Relationship-based scoping
Create Account	**Allowed**	**Allowed**	**Allowed**	Available to authenticated reps
Delete Account	**Allowed**	**Forbidden (403)**	**Forbidden (403)**	check_can_delete_account (Owner only)
View Contacts	**All Contacts**	**Team Contacts**	Rep Contacts	Parent account scoping
Create Contact	**Any Account**	**Team Accounts**	Rep Accounts	ContactService.create_contact
Delete Contact	**Any Contact**	**Team Contacts Only**	**Forbidden (403)**	check_can_delete_contact
View Deals (Kanban)	**All Deals**	**Team + Own**	Assigned Deals	COALESCE(d.owner_id, o.owner_id)
Create Deal	**Allowed**	**Allowed**	**Allowed**	DealService.create_deal
Move Deal Stage	**Any Deal**	**Team Deals**	**Assigned Deals**	Kanban stage advancement
Discount ≤ 10%	**Allowed**	**Allowed**	**Allowed**	Standard rep threshold
Discount 11% - 30%	**Allowed**	**Allowed**	**Requires Approval**	check_can_apply_discount
Discount > 30%	**Allowed**	**Requires Owner**	**Forbidden**	Strict Owner approval
Approve Discounts	**Any Amount**	**Up to 30%**	**Forbidden**	check_can_approve_discount
Delete Deal	**Any Deal**	**Team Deals Only**	**Forbidden (403)**	check_can_delete_deal
Assign Task to Others	**To Any Rep**	**To Team Reps**	Only to Self	check_can_assign_task
Delete Task	**Any Task**	**Created by Self**	**Created by Self**	check_can_delete_task
Manage Campaigns	**Full Access**	**Full Access**	**View Only (403)**	check_can_manage_campaign
Custom Report Builder	**Allowed**	**Allowed**	**Forbidden (403)**	check_can_build_reports
Analytics Dashboard	**Org-wide ($465k)**	**Team Subtotals**	Personal Pipeline	SQL role-scoped aggregates

4. Table-Level CRUD Permissions by Role

Summary of direct database operations (Create, Read, Update, Delete) and boundary filters across all 32 tables:

Table Name	Owner (Director)	Manager	Salesperson (User)	Scoping Boundary & Defense Policy
organizations	CRUD	R	R	Tenant boundary isolation
roles	CRUD	R	R	Tenant-scoped role definitions
permissions	R	R	R	Static capability catalog
role_permissions	CRUD	R	R	Role assignment mapping
users	CRUD	RU (Team reps)	RU (Own profile)	Hierarchical reporting tree
teams	CRUD	R	R	Manager assigned teams
team_members	CRUD	CRUD (Own team)	R	Team membership mapping
campaigns	CRUD	CRUD	R (View only)	check_can_manage_campaign
campaign_members	CRUD	CRUD	R	Campaign enrollment
campaign_responses	CRUD	CRUD	R	Ingestion pipeline
leads	CRUD	CRU + D (Team)	CRU (Assigned only)	leads.assigned_to = user_id
lead_signals	CR	CR	CR	Linked to accessible lead
lead_conversions	CR	CR	CR	Lead conversion audit trail
accounts	CRUD	CRU (No Delete)	CRU (No Delete)	check_can_delete_account (Owner only)
contacts	CRUD	CRU + D (Team)	CRU (No Delete)	check_can_delete_contact (Owner/Mgr)
opportunities	CRUD	CRU + D (Team)	CRU (Assigned only)	opportunities.owner_id = user_id
deals	CRUD	CRU + D (Team)	CRU (Assigned only)	deals.owner_id = user_id
products	CRUD	R	R	Central product price book
deal_products	CRUD	CRUD (Team deals)	CRU (Assigned deals)	Linked to accessible deal
sales_targets	CRUD	CRU (Team reps)	R (Self quota)	assigned_to_user_id = user_id
activities	CRUD	CRUD (Team tasks)	CRU + D (Self tasks)	activities.user_id = user_id
activity_participants	CRUD	CRUD	CRU	Linked to accessible activity
notes	CRUD	CRUD	CRU	Polymorphic entity linkage
invoices	CRUD	CRU	R (Assigned deals)	Finance and deal linking
invoice_items	CRUD	CRU	R	Linked to accessible invoice
payments	CRUD	CR	R	Payment remittance records
documents	CRUD	CRU	CRU (Self uploads)	documents.uploaded_by = user_id
document_relations	CRUD	CRUD	CR	Entity document linkage
document_versions	CRUD	CR	CR	Revision file storage
audit_logs	R (Read-only)	None	None	Immutable compliance trail
outbox_events	System	System	System	Transactional event outbox
idempotency_keys	System	System	System	API Gateway cache

5. Entity Relationships Specification (1:N and N:M)

Complete topology of parent-child (1:N) foreign keys and many-to-many (N:M) junction mappings:

A. Core 1 : N (One-to-Many) Relationships

Parent (1)	Child (N)	Foreign Key Column	On Delete	Business Meaning & Functional Cascade
organizations	roles	organization_id	CASCADE	Tenant defines multiple custom security roles
organizations	users	organization_id	CASCADE	Tenant employs multiple sales representatives and directors
organizations	teams	organization_id	CASCADE	Tenant organizes multiple regional sales teams
organizations	leads	organization_id	CASCADE	Tenant owns multiple inbound and qualified leads
organizations	accounts	organization_id	CASCADE	Tenant manages multiple client enterprise accounts
organizations	deals	organization_id	CASCADE	Tenant tracks multiple Kanban pipeline deals
organizations	campaigns	organization_id	CASCADE	Tenant runs multiple outbound marketing campaigns
organizations	invoices	organization_id	CASCADE	Tenant issues multiple commercial invoices
roles	users	role_id	SET NULL	Role assigned to multiple user accounts
users	users (self)	manager_id	SET NULL	Sales Manager supervises multiple sales representatives
users	teams	manager_id	SET NULL	Manager appointed to lead a sales pod/team
teams	team_members	team_id	CASCADE	Sales team has multiple member assignments
users	team_members	user_id	CASCADE	Sales rep enrolled across team assignments
campaigns	leads	campaign_id	SET NULL	Marketing campaign generates multiple leads
campaigns	campaign_members	campaign_id	CASCADE	Campaign enrolls multiple prospective contacts
campaign_members	campaign_responses	campaign_member_id	CASCADE	Enrolled member submits multiple replies / clicks
leads	lead_signals	lead_id	CASCADE	Lead accumulates multiple intent & behavioural signals
leads	lead_conversions	lead_id	CASCADE	Lead records historical conversion into an account
users	leads	assigned_to	SET NULL	Salesperson assigned multiple leads
accounts	contacts	account_id	CASCADE	Enterprise account employs multiple stakeholder contacts
accounts	opportunities	account_id	CASCADE	Enterprise account has multiple pipeline opportunities
accounts	deals	account_id	CASCADE	Enterprise account linked to multiple Kanban deals
accounts	invoices	account_id	CASCADE	Enterprise account billed across multiple invoices
contacts	opportunities	contact_id	SET NULL	Buyer contact is primary stakeholder on multiple deals
opportunities	deals	opportunity_id	SET NULL	Pipeline opportunity tracked by an active deal
users	opportunities	owner_id	SET NULL	Salesperson owns multiple pipeline opportunities
users	deals	owner_id	SET NULL	Salesperson owns and advances multiple Kanban deals
deals	deal_products	deal_id	CASCADE	Deal contains multiple line-item product purchases
products	deal_products	product_id	RESTRICT	Catalog item sold across multiple deal line items
deals	invoices	deal_id	SET NULL	Won deal billed across milestone invoices
invoices	invoice_items	invoice_id	CASCADE	Invoice contains multiple line items
invoices	payments	invoice_id	CASCADE	Invoice settled across multiple payments
users	sales_targets	assigned_to_user_id	CASCADE	Sales rep receives multiple quarterly targets
users	activities	user_id	SET NULL	Sales rep assigned multiple tasks and calls
activities	activity_participants	activity_id	CASCADE	Meeting activity has multiple attendees
documents	document_versions	document_id	CASCADE	Contract document maintains multiple revisions

B. N : M (Many-to-Many via Join Tables) Relationships

Entity A	Entity B	Intermediate Join Table	FK Column A	FK Column B	Cardinality & Functional Meaning
roles	permissions	role_permissions	role_id	permission_id	Roles hold many permissions; permissions map to many roles
teams	users	team_members	team_id	user_id	Teams have multiple reps; reps can belong to multiple pods
deals	products	deal_products	deal_id	product_id	Deals sell many products; products appear in many deals
campaigns	leads	campaign_members	campaign_id	lead_id	Campaigns target many leads; leads receive many campaigns
campaigns	contacts	campaign_members	campaign_id	contact_id	Campaigns target many contacts; contacts receive many campaigns
activities	users	activity_participants	activity_id	user_id	Activities host multiple reps; reps attend multiple activities
activities	contacts	activity_participants	activity_id	contact_id	Activities host multiple contacts; contacts attend many meetings

C. Polymorphic 1 : N Generic Relationships

Parent Entities (Any)	Dependent Table (N)	Discriminator Column	Foreign ID Column	Supported Parent Models
Leads, Deals, Accounts, Contacts	notes	related_to_type	related_to_id	'lead', 'deal', 'account', 'contact'
Deals, Accounts, Leads, Opportunities	document_relations	related_to_type	related_to_id	'deal', 'account', 'lead', 'opportunity'